

Executive Summary

Key Observations

1. Organic Banana as a Core Basket Item

We observed that Bag of Organic Bananas appears in 12.39% of all transactions, making it one of the most central products in the purchase network.

Several strong associations were identified with this item. For example:

- *(Organic Cucumber, Organic Raspberries) → Bag of Organic Bananas*
 - Confidence: 70.6%
 - Lift: 5.70
- *(Organic Hass Avocado, Organic Raspberries) → Bag of Organic Bananas*
 - Confidence: 60.0%
 - Lift: 4.84

These rules indicate that customers purchasing combinations of organic produce are 4–5× more likely to also purchase organic bananas compared to random chance.

Recommendation

Create a “Healthy Organic Bundle” including:

- Organic Bananas
- Organic Avocado
- Organic Raspberries

Promote this bundle in both in-store displays and online recommendations.

Fruit Basket Purchasing Pattern

Another strong pattern was observed around Banana purchases (15.1% support).

Examples include:

- *(Strawberries, Honeycrisp Apple) → Banana*
 - Confidence: 70.6%
 - Lift: 4.67
- *(Almond Breeze Original Almond Milk) → Banana*

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- Confidence: 57.1%
- Lift: 3.78

These patterns suggest bananas are frequently purchased alongside fruit and breakfast-related products.

Recommendation

Introduce a “Breakfast Fruit Pack” containing:

- Bananas
- Strawberries
- Honeycrisp Apples

Position this bundle near breakfast foods or promote it in weekly specials.

Organic Produce Cluster

Multiple rules link combinations of organic vegetables to Bag of Organic Bananas, including:

- *(Carrots, Organic Yellow Onion) → Bag of Organic Bananas*
 - Confidence: 55.6%
 - Lift: 4.48
- *(Organic Celery, Organic Yellow Onion) → Bag of Organic Bananas*
 - Confidence: 55.0%
 - Lift: 4.44

This indicates the presence of an organic produce purchasing cluster.

Recommendation

Place organic bananas near organic vegetables in store layouts and feature them together in “Organic Produce” promotional displays.

Overall Insight

The analysis reveals that banana (both regular and organic) act as hub products within the shopping network, linking multiple purchasing clusters including fresh fruit, organic produce, and breakfast items.

Strategic Recommendation

Leverage bananas as a cross-selling anchor product in promotions, store layout decisions, and recommendation systems to increase average basket size and overall revenue.

Expected Business Impact

The association rule analysis identifies high-confidence product relationships ($\geq 50\%$) with lift values between 3.3 and 5.7, indicating strong purchasing dependencies between several organic produce items and bananas.

By leveraging these relationships through product bundling, cross-promotion, and store layout optimization, retailers can increase the likelihood of multi-item purchases within the same basket.

Implementing the recommended bundles and adjacency placements is expected to:

- Increase Average Order Value (AOV) by encouraging complementary purchases
- Improve product discovery for related organic items
- Strengthen cross-selling effectiveness within the fresh produce category

Given that bananas and organic bananas act as central hub products in the transaction network, using them as anchor items for promotions and recommendations can significantly improve basket size and overall retail revenue.
